

Chapter 1

Introduction to IS

(Information System)

Basic IS Concepts

What is an IS ?

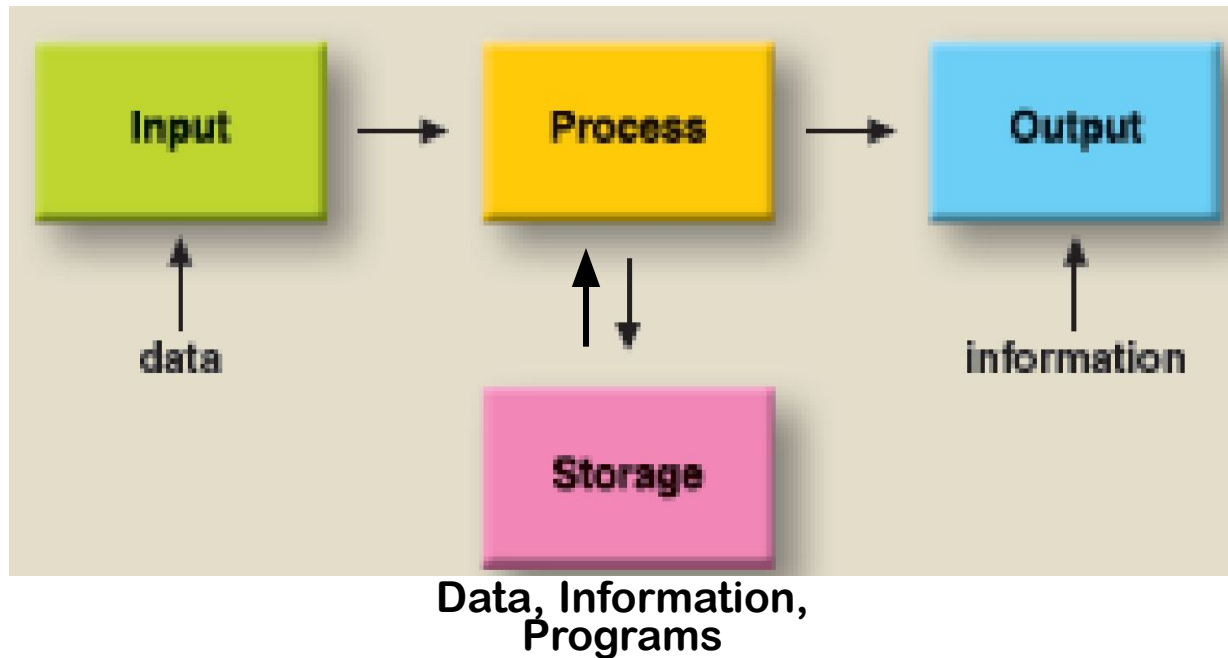
- **Components:**

- **Hardware** : servers, clients, network infrastructure
- **Software** : operating systems, DBMS and applications
- **Process** : consists of **activities** or tasks supported by the applications eg. create PO, make payments, receive goods
- **Data stored** : files or tables (transactions or masters)
- **People** : end-users, IT/IS personnels

- **Purpose :**

- To capture, store, input, process data and to produce *information* to help operate, support and manage organization.

Functions of Information System



- 4 Functions performed by IS (above)
 - **Input** : accepts raw **data** from outside the system
 - **Process** : processes the raw data into information
 - **Output** : produces the results of the process (**information**)
 - **Storage** : stores *data, information, programs*

Example 1 - Inventory Control System (Receiving Goods Module)

- **Input function** – Order and delivery (supplier) details (eg. PO No, item code, quantities)
- **Process function** – process the input data (ie. goods received data) and update the quantities in the inventory data store etc.
- **Output function** – the user (eg. store manager) retrieve information about the inventory details.
- **Storage function** – data (eg. goods received data) and information (updated inventory balance) stored for future use.

5 Components of an Information System

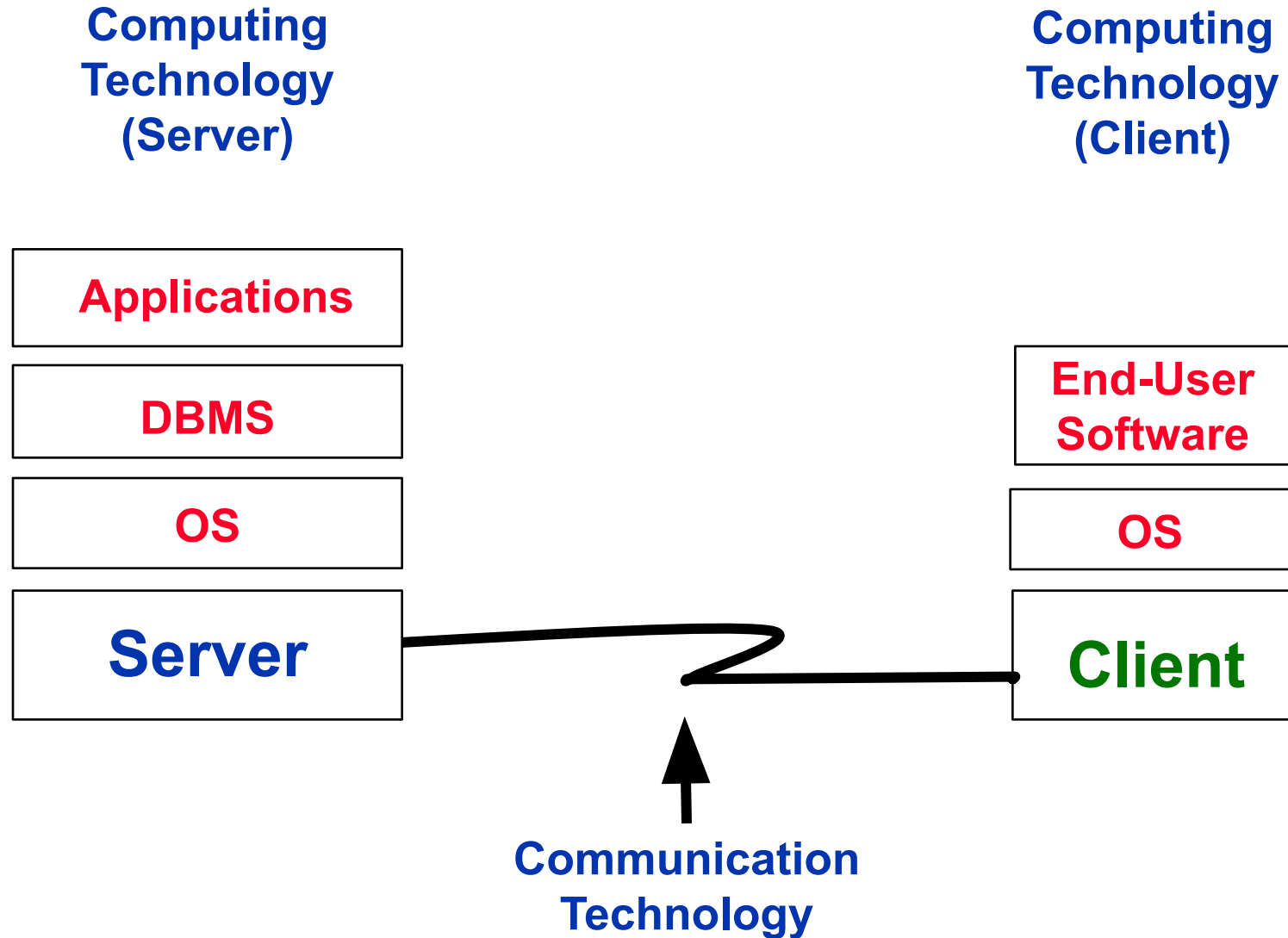
1. Hardware
2. Software
3. Processes
4. Data Stored
5. People



1. Hardware

- **Servers** - (eg. Mini-computers, mainframes)
- **Clients** - (eg. PC, smartphone, laptops, ATM etc)
- **Network Infrastructure** - (eg. wire, modems, hubs, routers, switches etc).
- **Peripherals** - (input, output, storage devices) eg. digital cameras, microphones, printers, readers, sensors, scanners etc

ICT Technologies (Hardware and Software)



2. Software

- **Operating System (OS)** – manages both the computer hardware and software resources of a computer system eg. Linux, Unix, OS/400, Network OS.
- **Database Management System (DBMS)** – manage data stored eg Oracle, DB2, SyBase.
- **Software Development Tools** – support programmers in developing programs eg. translators (eg compilers), tools (eg. CASE tools) and programming languages (eg. Python, C)
- **Application Systems** – supports the running of the various business processes eg. sales, purchasing, warehousing, manufacturing, HR systems.

3. Processes

- **Processes** – contain *steps* that tell people *how to use and operate* the system to perform tasks/activities involving *data*.
- **Example** – Inventory Control System needs procedures :
 - How to enter **input** data
 - How to **process** the data into information
 - How to **output** information from the system.
 - How to **operate** the hardware and software.

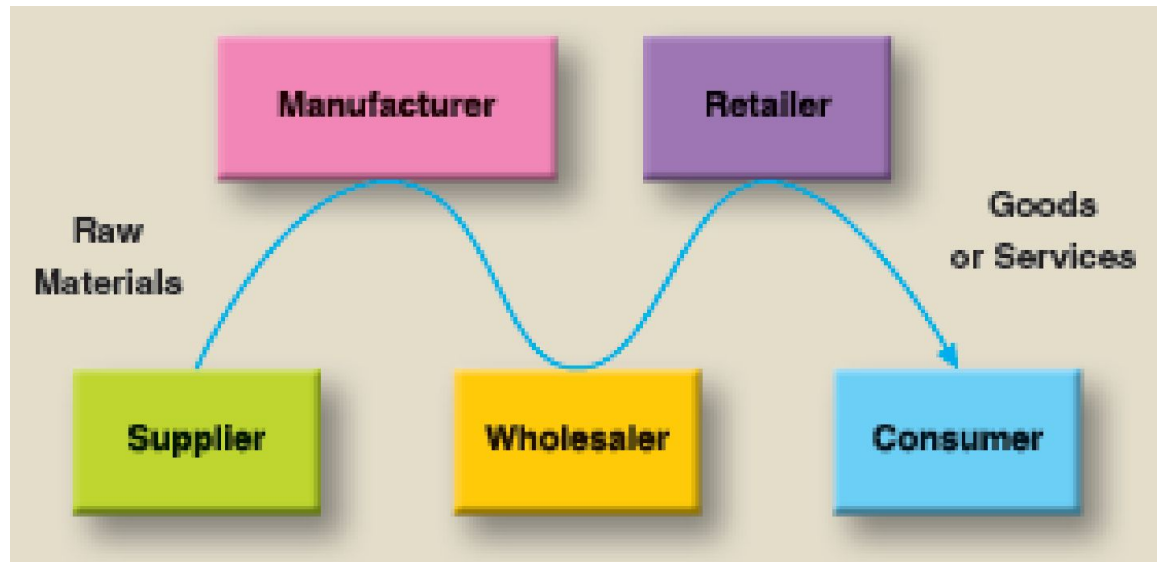
4. Data Stored

- Data that is **processed** by the software (eg payroll), **retained** in the hardware (eg harddisk) and **used** by the people (eg end-users).
- **Example** - Inventory Control System, the stored data :
 - **Master File** (Stock File) eg Stock code, name, unit of measurement, quantity, unit price.
 - **Transaction File** (Goods Received File) eg. Goods Received number, date received etc.

5. People

- People are needed to run system.
 - *input* data (**Users**)
 - *receive* output information (**Management**)
 - *operate* the hardware and software (**Operation staff**)
- Example - Inventory Control System
 - **Users and Management** eg. *salesperson* (check stock), *purchasers* (check stock), *storehands* (receive and distribute goods), *store manager* (create inventory reports).
 - **Operation staff** - personnels operate the hardware and software in the system

Types of Organisations (Product)



- **Supplier** - Supply raw materials that flow through an organisation and finally to consumer.
- **Manufacturer** - Raw materials, eg. oil and iron, are manufactured into products such as gasoline and automobiles.
- **Wholesaler** - These products are sold to other businesses at wholesale – that is, in large quantities.
- **Retailer** - The products then are resold at retail – that is, in smaller quantities – to the final consumer.

Types of Organisation (Services)

- Services are **intangible** products.
- **Examples of services :**
 - Banking, insurance, logistics, consulting, delivery and repairs, maintenance
 - Government
 - **Purpose** : formulate laws and regulations
 - **Examples** : public safety, policing, fire protection
 - Non-Profit Organisations
 - **Purpose** : charitable by providing goods and services on voluntary basis
 - **Examples** : education, medical care, religious services, cultural activities

Business Functions of an Organisation



Business Functions - Examples

- Marketing Functions

- Determine **prices** of products and services
- **Advertising** and promotions of products and services
- Carry out **promotions**
- Carry out market **surveys** of customers' needs or change in tastes and trends
- Carry out market **intelligence** of competitors

- Sales Functions

- Answer **enquiries** from customers.
- Create **quotations** for customers.
- Accept **orders** from customers eg. create stock and customer credit status.
- **Collaborate** with other departments for delivery of

Business Functions - Examples

- **Production Functions**

- **Production planning** – deciding what (products) to produce, how much to produce and when to produce.
- **Capacity planning** – determining capacity requirements (machines and human resources).
- Determine **materials** requirements.
- **Monitoring** and control production progress.

- **Human Resource Functions**

- Assessment of staff **requirement**
- **Recruitment** of staff eg. evaluation and selection
- **Training and development** eg. create training plan, carry out training, record training, conduct training needs analysis
- **Payroll** eg calculate gross and net pay, create payslip, compute deductions.
- Staff **performance appraisal**

Business Functions - Examples

- **Accounting Functions**

- **Recording** business transaction entries (debits and credits)
- **Maintain** assets records (items the business owns eg. cash, equipment, and buildings)
- **Maintain** liabilities records (eg. items business owe others)
- **Reporting** – financial profitability (eg P&L) and financial *state* of business (eg. balance sheets)

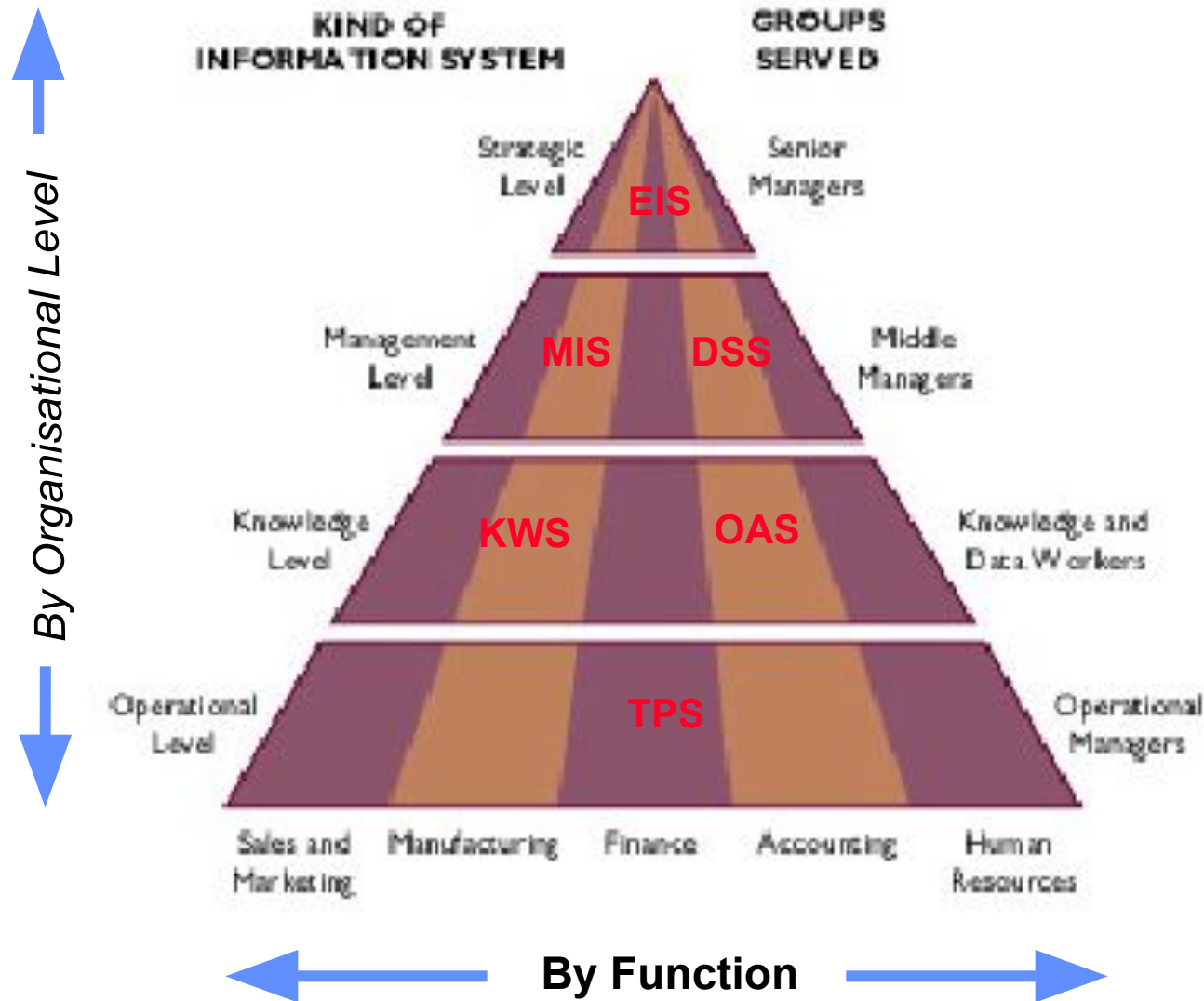
- **Finance Functions**

- **Carry** out budgeting
- **Obtaining** money needed by the business
 - Sale of goods and services eg. profits earned
 - Investments made by the business eg shares
 - Banks and other institutions eg. loans
- Plan and control the **use** of money (cash flows).

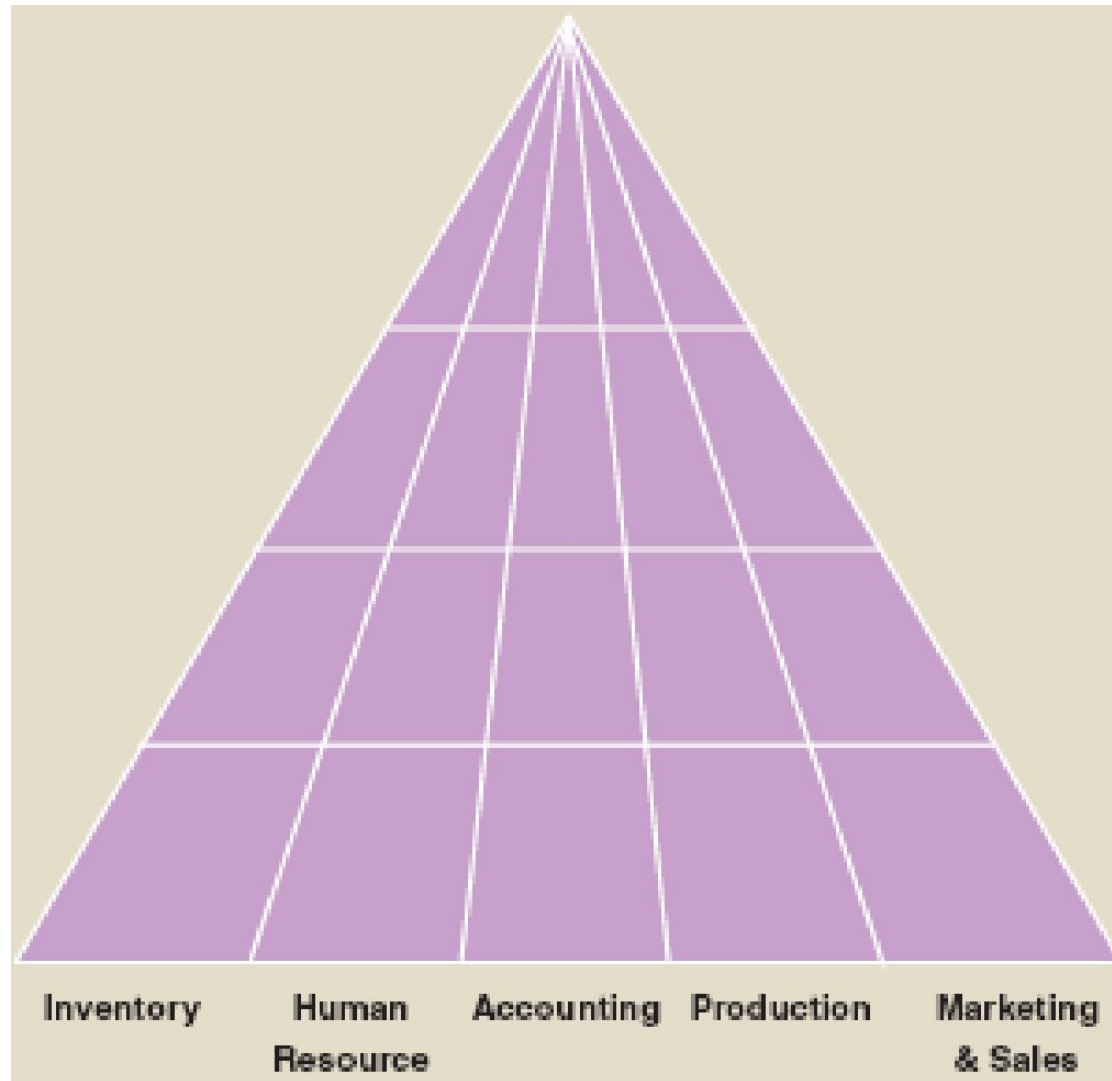
Business Functions - Examples

- **Information Technology Functions**
 - Analysis and Design of users' needs
 - Developing software
 - Buy packages
 - Implementing eg. training, data migration
 - Operation - assisting other functional areas (depts) on using IT.
 - Maintenance

Information Systems Used (categories)



ISs by Functional Areas



Examples of ISs by Functions

- **Purchasing System (buy)**
- **Sales System (sell)**
- **Inventory System (store)**
- **Production System (make)**
- **Accounting System (track)**
- **Finance System (track)**
- **Marketing System (promote)**
- **HR System (manage)**

Exercise for Students

- List any 4 functional systems for an **airline**
- List any 4 functional systems for a **manufacturing** company
- From the list indicate which systems are 'core' and which are 'non-core'.

Solutions

- **List any 4 functional systems for an airline**
 - Air-ticket reservation system (core)
 - Check-in system (core)
 - Flight scheduling system (core)
 - Plane maintenance system (core)
 - Accounting system (non-core)
 - HR system (non-core)
- **List any 4 functional systems for a manufacturing company**
 - Production system (core)
 - Sales system (core)
 - Purchasing system (core)
 - Inventory system (core)
 - CAD system (core)
 - Accounting system (non-core)

Enterprise Systems (**Characteristics**)

- **System.** Single large system consisting of many modules eg. ERP, CRM, SCM systems.
- **Integrate.** ES integrates business *processes*, *databases*, and *functional areas* (internally and externally). Allows data flows (vertically and horizontally).
- **Data Storage.** ES provides a *centralized database storage* for all users (intra or inter-companies).
- **Data Sharing.** ES allows users to share information *seamlessly* no matter where the data is located and used for decision making eg. inventory data shared with sales and purchasing staff.
- **Purpose.** ES *automates* business *processes* of the entire enterprise eg. order-to-cash process.

Types of Enterprise Systems

- Enterprise Resource Planning (**ERP**) system
- Customer Relationship Management (**CRM**) System
- Supply Chain Management (**SCM**) system

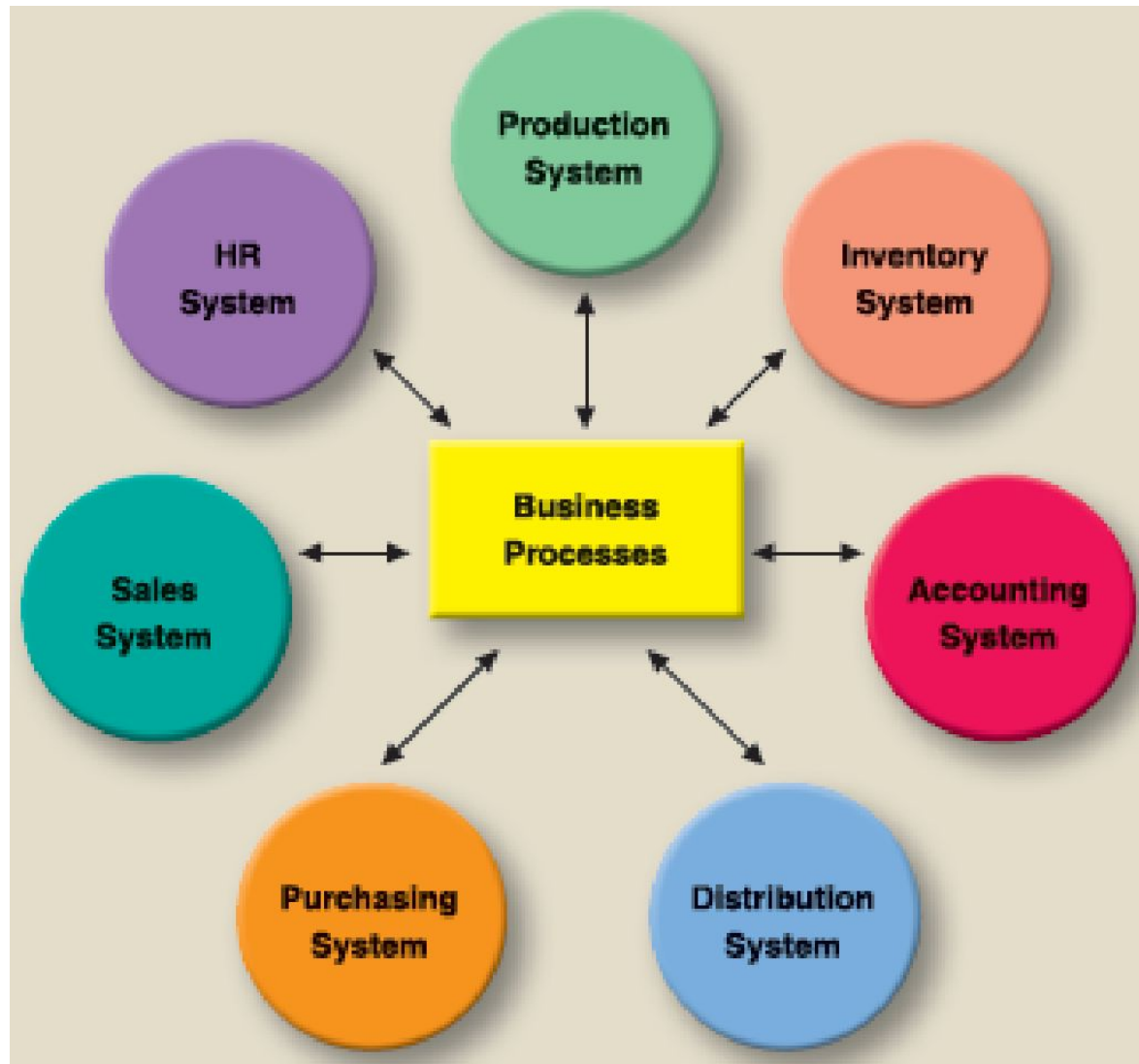
What is ERP ?



Diagram : Adapted from Laudon and Laudon

ERP **integrate** the key business processes of an entire firm into a single large software system that allows information to **flow** seamlessly throughout the organisation. These systems may **include** (connect) transactions with customers and vendors

Integrated Processes in ERP

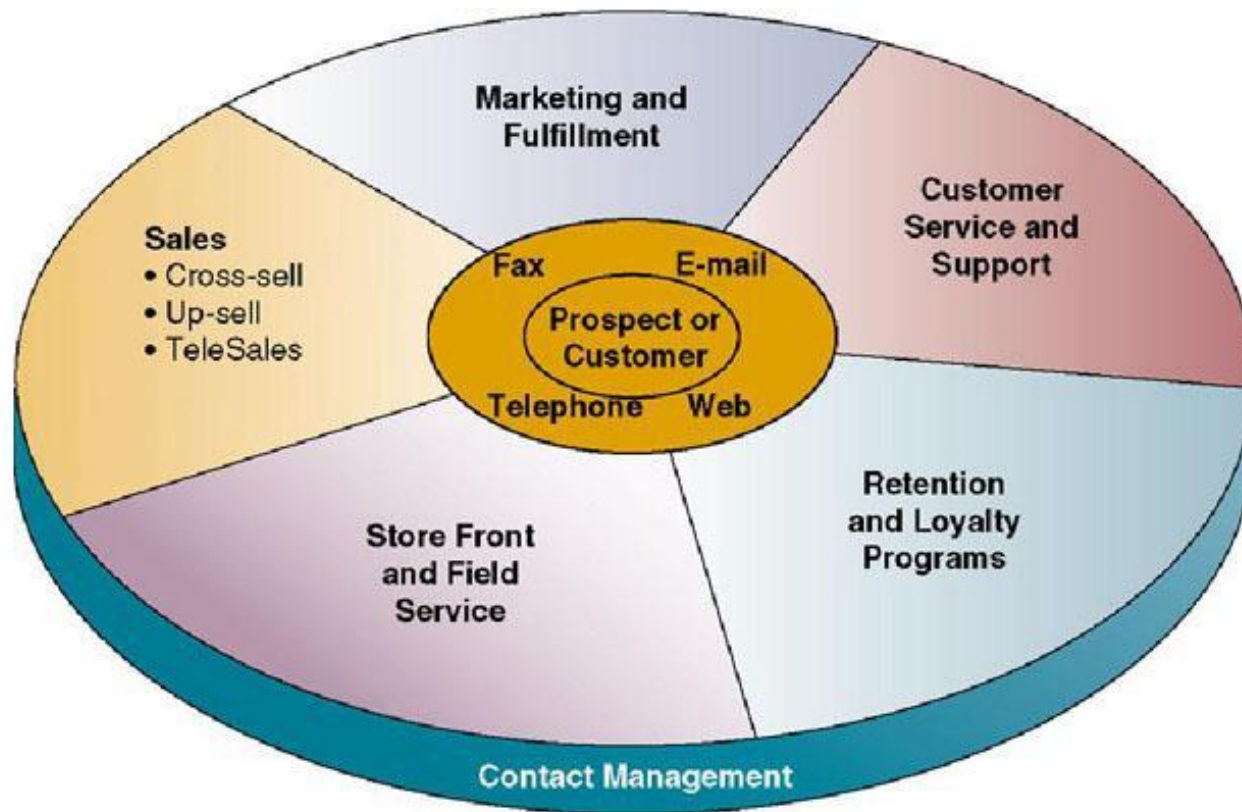


What is CRM ?

- Uses **technology** (data mining) to look (analyse) at customers from a multifaceted perspectives eg. buy patterns, habits, income.
- Uses a set of integrated **applications** to address all issues about customers *relationship* in: *sales, marketing and service*.

Typical Modules of CRM

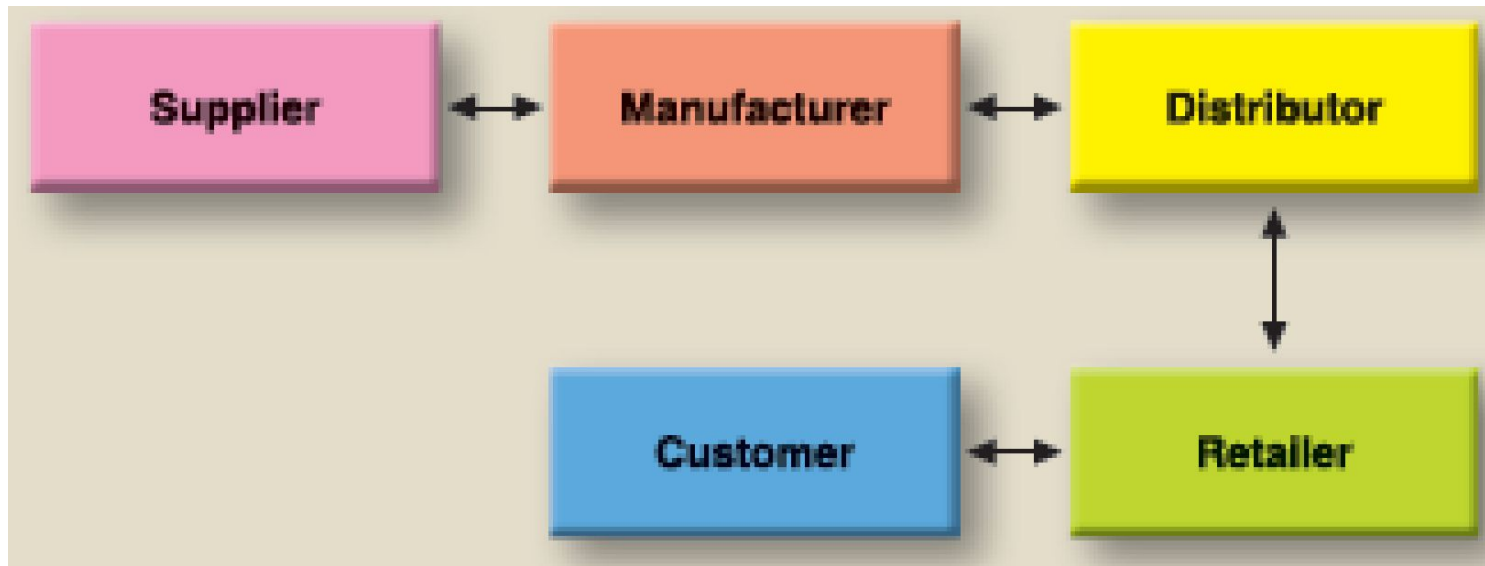
- CRM systems consists of a family of software modules that perform business activities involved mainly in **front office** processes.



Typical Modules of CRM

- **Marketing.** CRM automates various tasks related to *pre-selling* activities eg. qualifying leads, scheduling sales contacts with customers, and providing information.
- **Sales.** CRM helps to answer enquiries, create quotations and accept orders from customers. Carry out potential up-selling and cross-selling tasks.
- **Service.** CRM helps in *post-selling* activities to provide good customer service eg. creation of service requests, assign qualified personnel.

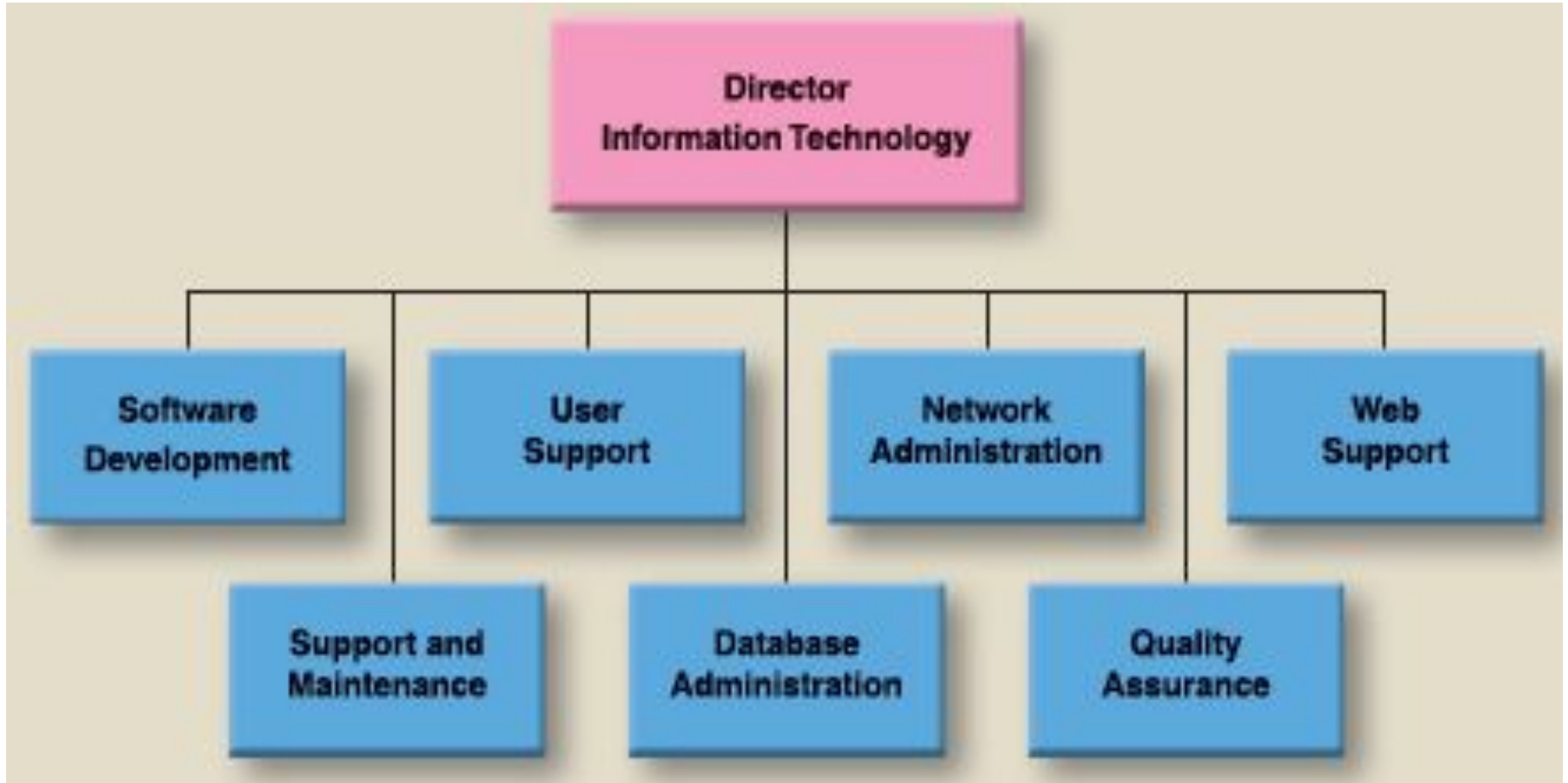
Supply Chain Management (SCM)



- **Purpose** – to deliver product more rapidly to customers, to reduce time, redundant effort, inventory costs
- **Linkage of activities** : buying, making, moving product
- **Integrates** - supplier, manufacturer distributor, retailer and customer
- **Network of facilities** - procuring, transforming and distributing products to customers.

IT Department

IT Department (Functional Areas)

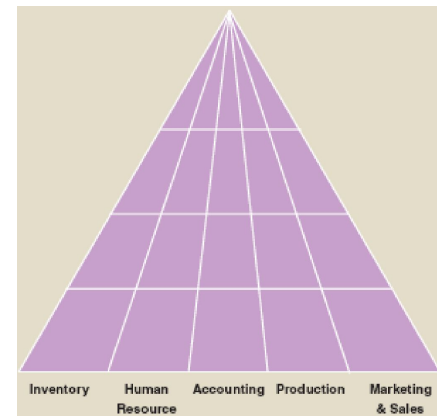


IT Department (Functional Areas)

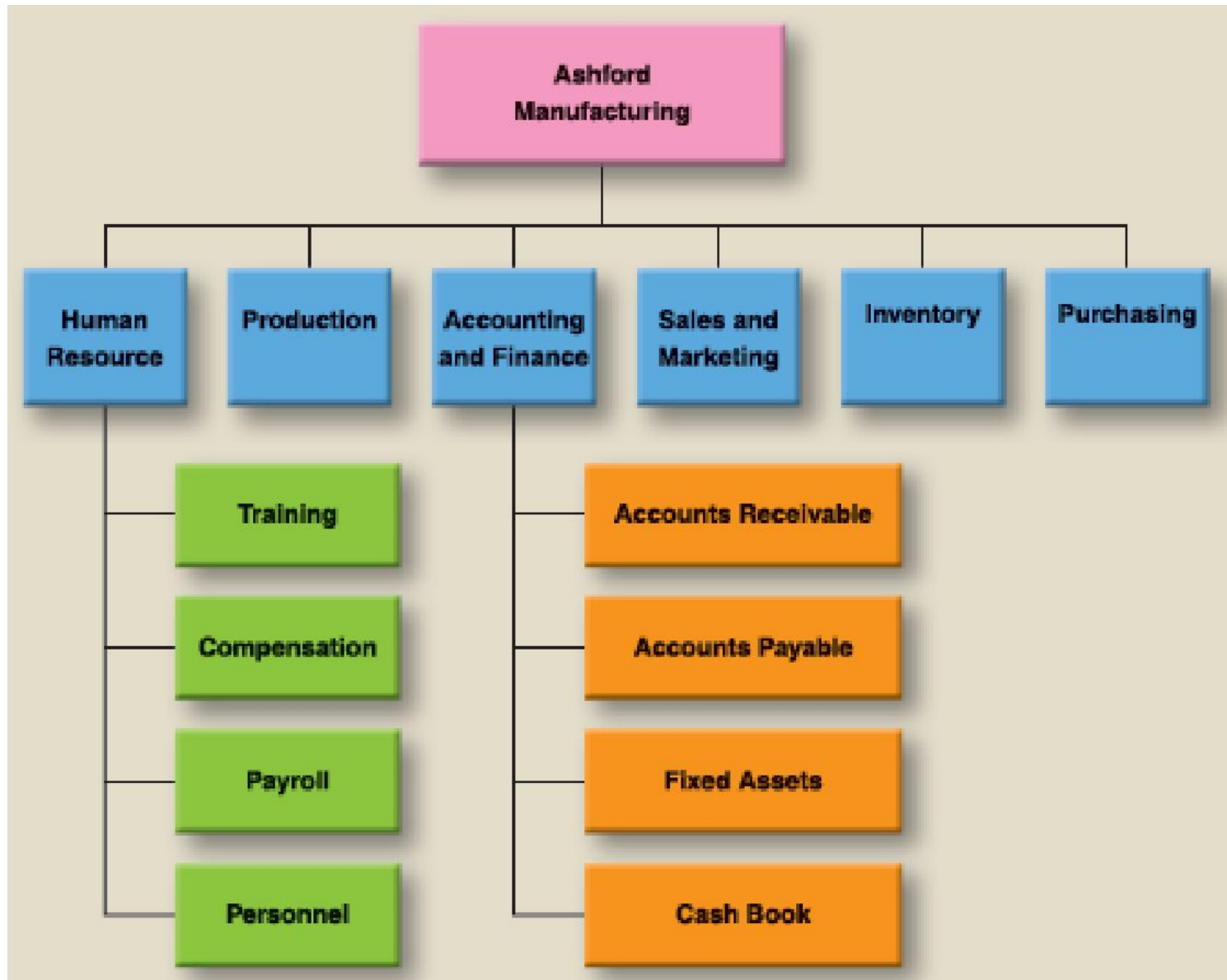
1. **Software Development** – teams use tools to develop programs
2. **Quality Assurance (QA)** – ensure quality of software developed, monitor changes, control quality standards
3. **Web Support** – support Intranets, Extranet operations, EC, develop websites
4. **Database Administration** – database design, backup, access and security control
5. **Network Administration** – maintenance and support eg. control user access, configuring, monitoring
6. **Operation Support** – eg training, operating, help desks, trouble shooting.
7. **Maintenance** – for both hardware and software eg. PCs, networks, ISs

Organisation of a Business

- **Organisation Structure** – *employees* are grouped based on departmental **functions** eg. Accounting, Sales etc.
- **Organisation Levels** – *employees positions* are grouped into vertical levels for **authority** control eg. HQ, subsidiaries, business units, branches, departments, product lines, regions, workgroups



Example of Organisation Chart



SDLC

Brief Introduction to SDLC

Phase 1

System Request

System Planning

Phase 2

System Analysis

Preliminary Investigation Report

Phase 3

System Design

System Requirements Specification

Phase 4

System Development

System Design Specification

Phase 5

System Implementation

Completed System

Phase 6

Operation and Maintenance

Operational System

